

DEV_IMPL_GUIDE - DD: BIL-02-GEN-01 — Invoice Generation

1. Purpose

This guide explains how to implement DD_BIL-02-GEN-01-Invoice_Generation-v1.3.md as executable SOPHIX behavior.

Use it with:

- DD_BIL-02-GEN-01-Invoice_Generation-v1.3.md
- DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0.md when this DD participates in subscription workflow operations.
- DD_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md when this DD references worker topics.

2. Runtime Actors

Runtime component	Responsibility
API pod / module service	Receives requests, validates, loads authoritative state, starts commands.
Owning module database	Stores tables/configuration owned by this DD.
Reference modules	Provide data through approved APIs/client wrappers.
Camunda	Runs BPMN process when this DD is workflow-driven.
Worker apps	Execute logical worker topics.
Drools	Evaluates packages when rules are declared.
Kafka/outbox	Publishes success/failure and integration events.

3. Before Implementation

Confirm these contracts from the DD:

Contract	Value
Endpoint scan reference	GET /api/bulk-reversals/preview?...
BPMN/process key	configured BPMN process
Drools package	configured Drools package
Tables	No CREATE TABLE block declared
Events	InvoiceCancelled, InvoiceGenerationContext, InvoiceGenerator, InvoiceGeneratorRegistry, InvoiceIssued, InvoiceWriter

4. Execution Flow

1. Apply migrations and seed configuration/catalog rows.
2. Implement the module service/repository layer around the tables and invariants.
3. Implement public/internal APIs from the DD.
4. Implement rule evaluation if the DD defines Drools packages.
5. Implement event production/consumption through outbox where the DD defines events.
6. Add integration tests for table constraints, API validation, and event payloads.

5. API And Command Handling

Endpoints found in the DD:

- GET /api/bulk-reversals/preview?...
- GET /api/bulk-reversals/{id}

- GET /api/bulk-reversals/{id}/failures
- GET /api/bulk-reversals?operator={code}&status={s}
- GET /api/generation-failures
- GET /api/generation-failures/{id}
- GET /api/invoices/{id}
- GET /api/invoices/{id}/pdf
- GET /api/invoices/{id}/status
- POST /api/billing/charges/compute
- POST /api/bulk-reversals
- POST /api/bulk-reversals/{id}/approve
- POST /api/bulk-reversals/{id}/cancel
- POST /api/credit-notes
- POST /api/generation-failures/{id}/resolve-no-action
- POST /api/generation-failures/{id}/retry
- POST /api/invoices/cycle/generate
- POST /api/invoices/manual
- POST /api/invoices/oneoff
- POST /api/invoices/{id}/cancel

Implementation rules:

- Resolve actor user and role from auth context, not from public request body.
- Check idempotency before inserting a new operation or creating a side effect.
- Return the existing operation/result for the same idempotency key and same request hash.
- Return 409 IDEMPOTENCY_KEY_REUSED for the same key with a different request hash.
- Use transactions or unique constraints for concurrency-sensitive creation.

6. Data Loading

Load data in this order:

1. Authoritative aggregate state from the owning module.
2. Operator/module config from this DD's config tables.
3. Catalog/reference data from approved APIs or client wrappers.
4. Current in-flight operation state from the framework, where relevant.

Do not read current mutable state from cache for state-changing operations.

7. Drools Configuration

Packages found:

- No Drools package explicitly declared.

For each package:

- Define fact classes that mirror the request, current state, config, and reference data.
- Keep rule output as explicit decision fields.
- Unit-test accept and reject cases from the DD.
- BPMN should re-check critical rules after start if state may have changed.

8. BPMN And Worker Topics

Worker topics found:

- billing-engine
- billing-invoicing
- billing-invoicing-gen
- put-master-fields

Implementation rules:

- Model service tasks with `camunda:type="external"` and the exact topic string.
- Worker handler must validate required variables before calling downstream services.
- Store outputs as process variables with stable names.
- Use message correlation keys exactly as specified by the DD.
- Add compensation paths for each external side effect.

9. Events

Events found:

- InvoiceCancelled
- InvoiceGenerationContext
- InvoiceGenerator
- InvoiceGeneratorRegistry
- InvoiceIssued
- InvoiceWriter

Event rules:

- Emit success only after authoritative commit.
- Emit rejected/failure event only after compensation/final failure decision.
- Use the outbox table/dispatcher; do not publish directly inside the transaction boundary unless the platform standard allows it.

10. Smoke Tests

Add tests for:

- Happy path.
- Invalid role or trigger.
- Invalid state.
- Missing config/catalog row.
- Dependency failure.
- Retry/idempotency.
- Event payload correctness.
- Worker topic failure and compensation.

11. Common Mistakes

- Treating worker-topic names as shell commands.
- Querying another module's database directly.
- Reading current state from cache in a state-changing flow.

- Emitting success before final commit.
- Hiding manual recovery paths for partial external side effects.
- Creating one deployment per topic when a consolidated worker app would be easier to operate.